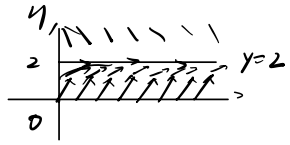
First order $\frac{dy}{dt} = f(t, y) =$ slope of y
slope of arrows

e.g. $\frac{dy}{dt} = 2 - y$

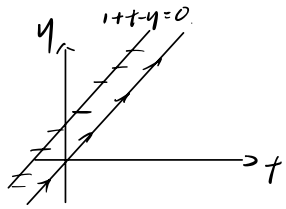
$y(t) = 2 + Ce^{-t}$



e.g. $\frac{dy}{dt} = 1 + t - y = f(t, y)$

$y(t) = t + Ce^{-t}$

$1 + t - y = 1 \Rightarrow y = t$



3.2. Source, Sink, Saddle

$y = c_1 e^{s_1 t} + c_2 e^{s_2 t}$

phase plane has axes $\boxed{y, y'}$!

$y' = c_1 s_1 e^{s_1 t} + c_2 s_2 e^{s_2 t}$

$y'' - 3y' + 2y = 0 \Rightarrow s^2 - 3s + 2 = 0$

$(s-1)(s-2) = 0$

$s = 1, s = 2$

$y = c_1 e^{t} + c_2 e^{2t}$

$y' = c_1 e^{t} + 2c_2 e^{2t}$

$s^2 + 3s + 2 = 0$

$(s+1)(s+2) = 0$

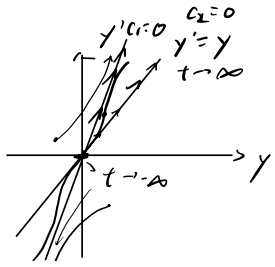
$s = -1, s = -2$

II

$c_1 = 0$

$y' = 2y$

$e^{t}\begin{pmatrix}1\\1\end{pmatrix} + \begin{pmatrix}1\\2\end{pmatrix}e^{2t}$



$t \to \infty$, approach e^{2t}

$t \to -\infty$ approach e^{t}

$\boxed{\text{phase}}$ saddle

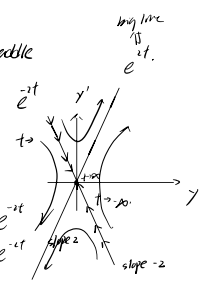
$y'' = 4y$

$s^2 = 4$

$s = \pm 2$

$y = c_1 e^{2t} + c_2 e^{-2t}$

$y' = 2c_1 e^{2t} - 2c_2 e^{-2t}$



$c_1 = 0$	$c_2 = 0$
$\boxed{y' = -2y}$	$y' = 2y$

$y = e^{-2t}$ ⊗ y正

$y' = -2e^{-2t}$

$t \to \boxed{-\infty}$

spirals & centers

$Ay'' + By' + Cy = 0$

$As^2 + Bs + C = 0$

$y = e^{st}$

$$\frac{-B \pm}{2}$$

Bc ⓑ

32 : spirals + centers

$Ay'' + By' + cy = 0 \Rightarrow s = a \pm iw$ for $B^2 < 4AC$

unstable

spiral source : $a > 0$, $B < 0$

center : $a = 0$, No damping

spiral sink : $a < 0$, underdamping

stable

$e^{(a+iw)t}$

Center : $y'' + w^2 y = 0$, $y = c_1 \cos wt + c_2 \sin wt$

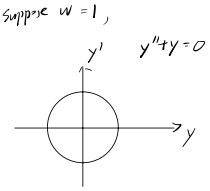
pure oscillation

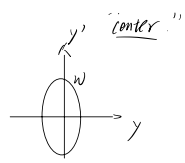
$y' = -c_1 w \sin(wt) + w c_2 \cos wt$

$y'' = -c_1 w^2 \cos(wt) - w^2 c_2 \sin(wt)$

suppose $w = 1$,

$y'' + y = 0$



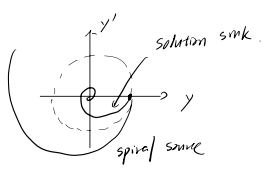


Spiral source / sink

$y'' \pm 2y' + 2y = 0$

$s^2 \pm 2s + 2 = 0$

$s = \frac{\pm 2 \pm 2i}{2} = \pm 1 \pm i$ $\begin{cases} 1 \pm i \\ -1 \pm i \end{cases}$



$e^{(-1 \pm i)t} = e^{-t}(c_1 \cos t + c_2 \sin t)$

$y'' + By' + Cy = 0 \Rightarrow \frac{d}{dt}\begin{bmatrix} y \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -c & -B \end{bmatrix}\begin{bmatrix} y \\ y' \end{bmatrix}$ s_1, s_2 或 λ_1, λ_2

companion matrix

$\begin{cases} s_1 + s_2 = -B \\ s_1 s_2 = C \end{cases}$ ⇒ 韦达

稳定 or not ? ⇒ s_1 & s_2 的实部是否小于 0.

$$\begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} = (a-\lambda)(d-\lambda) - bc = 0$$

$$\lambda^2 - (a+d)\lambda + ad - bc = 0$$

$$\lambda_1 + \lambda_2 = a + d = Tr(A)$$

$$\lambda_1 \cdot \lambda_2 = det(A)$$

3.3 Linearization at critical points.

$$f(y) = \underset{\overset{\shortparallel}{0}}{f(Y)} + (y-Y)\frac{df}{dy}(Y) = (y-Y)\frac{df}{dy}(Y)$$

$\frac{dy}{dt} \simeq (y-Y)\frac{df}{dy}(Y)$, 左侧加一常数.

$\frac{d(y-Y)}{dt} \simeq (y-Y)\frac{df}{dy}(Y)$, Let $\frac{df}{dy}(Y) = a$.

$\Rightarrow y - Y \simeq Ce^{at}$, if $a<0$, $y-Y \to 0$,

e.g. $\frac{dy}{dt} = 3y - y^2 = y(3-y)$

$Y=0$, $Y=3$.

$\frac{\partial f}{\partial y} = 3-2y$, $Y=3 \Rightarrow \frac{\partial f}{\partial y} = -3 < 0$ 稳.

$\lambda^2 + 1 = 0$

$\lambda = \pm i$

$\lambda = i$

$$\begin{pmatrix} -2-i & -1 \\ 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 \\ -2-i \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} + i\begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

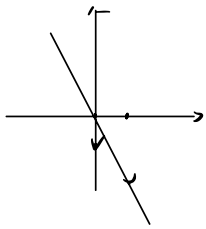
$$u = \cos t\begin{pmatrix} 1 \\ -2 \end{pmatrix} - \sin t\begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$v = \sin t\begin{pmatrix} 1 \\ -2 \end{pmatrix} + \cos\begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$\vec{x} = c_1 u_1 + c_2 u_2$$

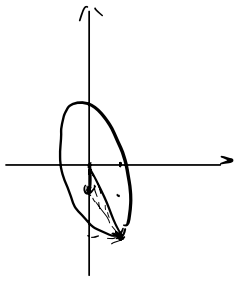
$$= (c_1\cos t + c_2 \sin t)\begin{pmatrix} 1 \\ -2 \end{pmatrix} + (-c_1 \sin t + c_2 \cos t)\begin{pmatrix} 0 \\ -1 \end{pmatrix}$$

$$= \begin{pmatrix} c_1\cos t + c_2 \sin t \\ (-2c_1 - c_2)\cos t + (c_1 - 2c_2)\sin t \end{pmatrix}$$



center.

$t=0$.

$$= \begin{pmatrix} 1 & 0 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} c_1 \cos t + c_2 \sin t \\ -c_1 \sin t + c_2 \cos t \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$